

# RAPPORT DE PROJET

## JAVA Système de Gestion d'École

CPI2D

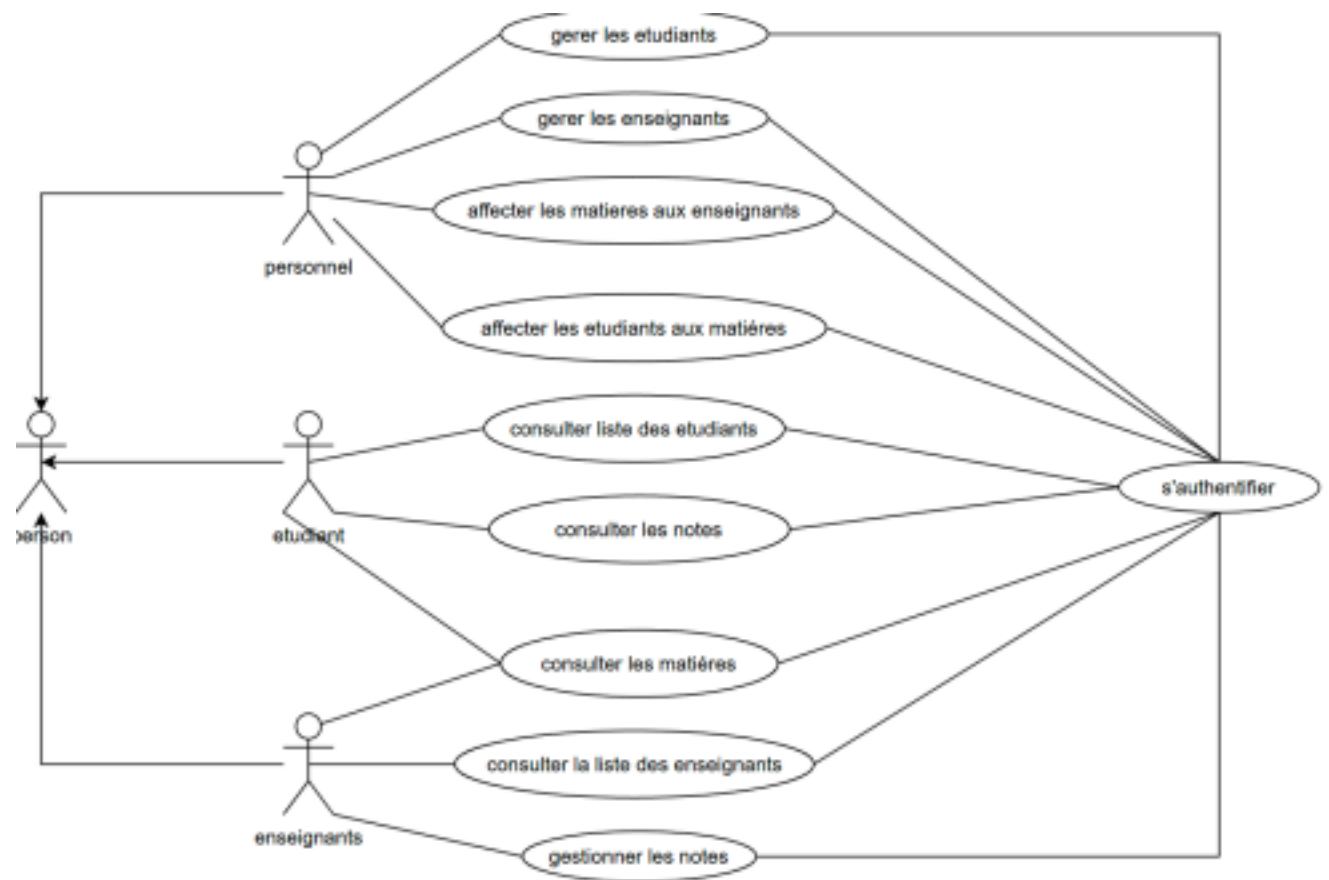
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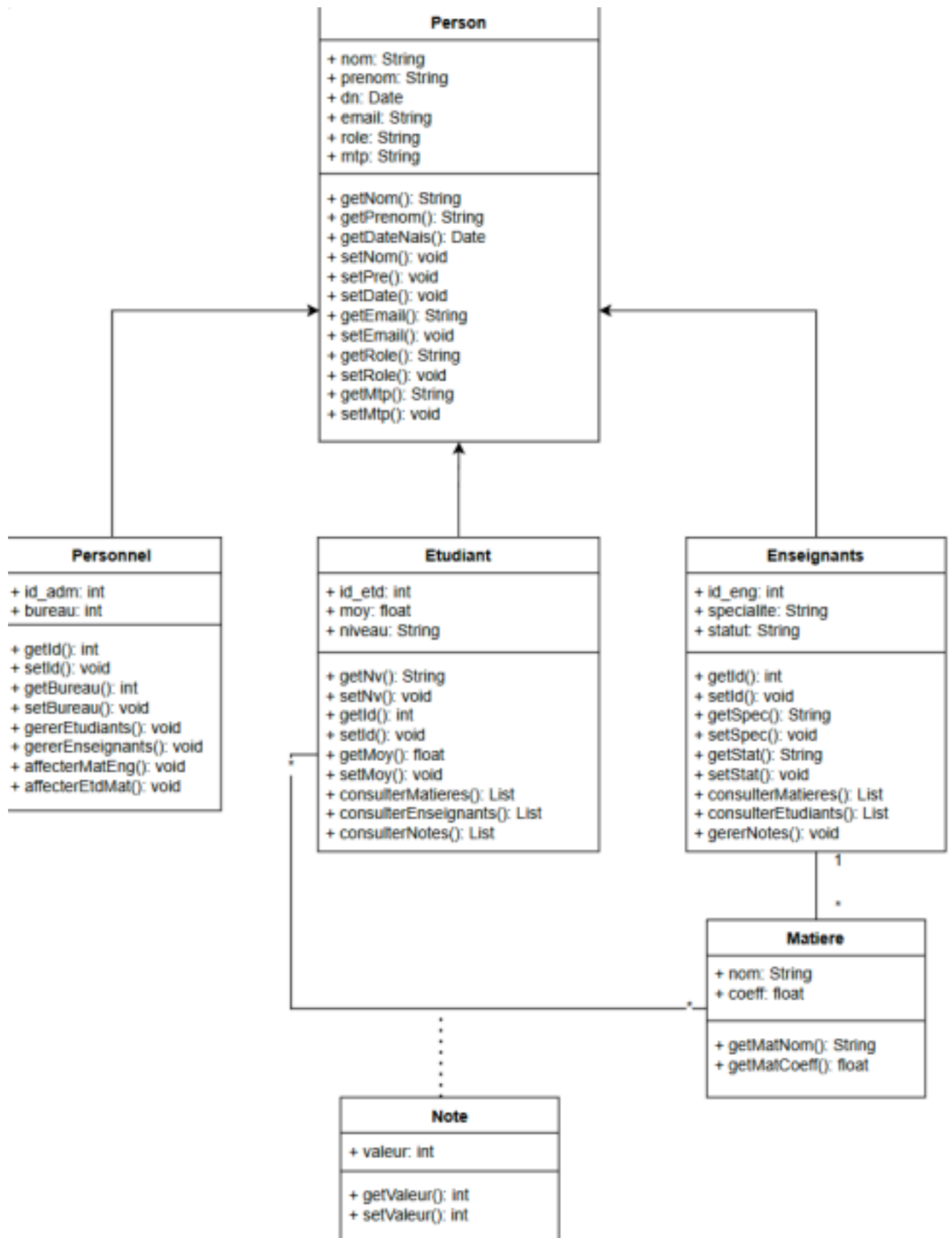
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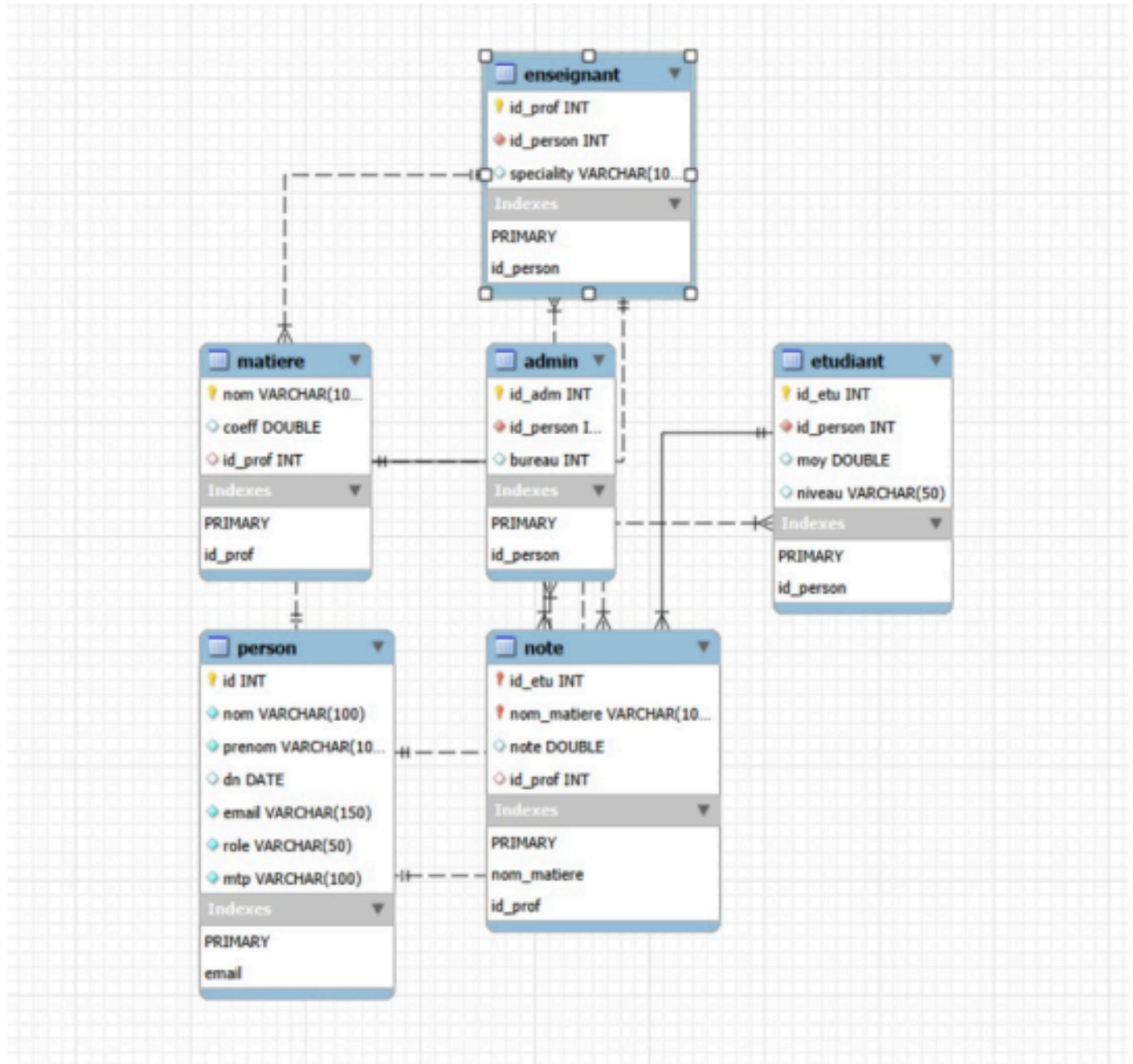
## Diagramme de Cas d'Utilisation



## Diagramme de Classe



## Schéma Relationnel



# Code Source : Couche Modèle

## Fichier : Person.java

```
package module;

public abstract class Person {
    private int id;
    private String nom;
    private String prenom;
    private String dn;
    private String email;
    private String role;
    private String mtp;

    public Person(int id, String nom, String prenom, String dn, String email, String role, String mtp) {
        this.id = id;
        this.nom = nom;
        this.prenom = prenom;
        this.dn = dn;
        this.email = email;
        this.role = role;
        this.mtp = mtp;
    }

    public Person() {

    }

    public int getId() {
        return id;
    }

    public void setId(int id) {
        this.id = id;
    }

    public String getNom() {
        return nom;
    }

    public void setNom(String nom) {
        this.nom = nom;
    }

    public String getPrenom() {
        return prenom;
    }

    public void setPrenom(String prenom) {
        this.prenom = prenom;
    }

    public String getDn() {
        return dn;
    }

    public void setDn(String dn) {
        this.dn = dn;
    }
}
```

```

public String getEmail() {
return email;
}

public void setEmail(String email) {
this.email = email;
}

public String getRole() {
return role;
}

public void setRole(String role) {
this.role = role;
}

public String getMtp() {
return mtp;
}

public void setMtp(String mtp) {
this.mtp = mtp;
}
}

```

## Fichier : Admin.java

```

package module;

public class Admin extends Person {
    private int id_adm,id_person,bureau;
    public Admin(int id_adm, int id_person, int bureau) {
        this.id_adm = id_adm;
        this.id_person = id_person;
        this.bureau = bureau;
    }
    public Admin(int id, String nom, String prenom, String dn, String email,
String role, String mtp, int id_adm, int id_person, int bureau) {
        super(id, nom, prenom, dn, email, role, mtp);
        this.id_adm = id_adm;
        this.id_person = id_person;
        this.bureau = bureau;
    }
    public int getId_adm() {
        return id_adm;
    }

    public void setId_adm(int id_adm) {
        this.id_adm = id_adm;
    }

    public int getId_person() {
        return id_person;
    }

    public void setId_person(int id_person) {
        this.id_person = id_person;
    }
}

```

```

public int getBureau() {
    return bureau;
}
public void setBureau(int bureau) {
    this.bureau = bureau;
}
}

```

## Fichier : Enseignant.java

```

package module;

public class Enseignant extends Person {
    private int id_prof;
    private int id_person;
    private String speciality;
    private int id_etd;
    private double note;

    public Enseignant(int id, String nom, String prenom, String dn, String
email, String role, String mtp, int id_prof, int id_person, String
speciality) {
        super(id, nom, prenom, dn, email, role, mtp);
        this.id_prof = id_prof;
        this.id_person = id_person;
        this.speciality = speciality;
    }
    public Enseignant(int id_etd, String speciality, double note, int
id_prof) {
        super();
        this.id_etd = id_etd;
        this.speciality = speciality;
        this.note = note;
        this.id_prof = id_prof;
    }
    public int getId_etd() {
        return id_etd;
    }

    public void setId_etd(int id_etd) {
        this.id_etd = id_etd;
    }

    public double getNote() {
        return note;
    }

    public void setNote(double note) {
        this.note = note;
    }

    public int getId_prof() {

```



```

        return id_prof;
    }

    public void setId_prof(int id_prof) {
        this.id_prof = id_prof;
    }

    public int getId_person() {
        return id_person;
    }

    public void setId_person(int id_peron) {
        this.id_person = id_peron;
    }

    public String getSpeciality() {
        return speciality;
    }

    public void setSpeciality(String speciality) {
        this.speciality = speciality;
    }
}

```

## Fichier : Etudiant.java

```

package module;

public class Etudiant extends Person{
    private int id_etu,id_person;
    private double moy;
    private String niveau;

    public Etudiant(int id, String nom, String prenom, String dn, String
email, String role, String mtp, int id_etu, int id_person, double moy,
String niveau) {
        super(id, nom, prenom, dn, email, role, mtp);
        this.id_etu = id_etu;
        this.id_person = id_person;
        this.moy = moy;
        this.niveau = niveau;
    }

    public Etudiant(int id_etu, int id_person, double moy, String niveau) {
        this.id_etu = id_etu;
        this.id_person = id_person;
        this.moy = moy;
        this.niveau = niveau;
    }

    public int getId_etu() {
        return id_etu;
    }

    public void setId_etu(int id_etu) {
        this.id_etu = id_etu;
    }

    public int getId_person() {

```

```

return id_person;
}

public void setId_person(int id_person) {
this.id_person = id_person;
}

public double getMoy() {
return moy;
}

public void setMoy(double moy) {
this.moy = moy;
}

public String getNiveau() {
return niveau;
}

public void setNiveau(String niveau) {
this.niveau = niveau;
}
}

```

### Fichier : Matiere.java

```

package module;

public class Matiere {
    private String nom;
    private double coeff;
    private int id_prof;
    public Matiere(String nom,
double coeff, int id_prof) {
this.nom = nom;
this.coeff = coeff;
this.id_prof = id_prof;
}
    public String getNom(){
return nom; }
    public double getCoeff(){
return coeff; }
    public int getId_prof(){
return id_prof; }
    public void setNom(String nom){
this.nom = nom; }
    public void setCoeff(double coeff){
this.coeff = coeff; }
    public void setId_prof(int id_prof){
this.id_prof = id_prof; }
}

```

### Fichier : Note.java

```

package module;

public class Note {
    private int id_etu;
    private String nom_matiere;
    private double valeur;
    private int id_prof;

```

```
    public Note(int id_etu, String nom_matiere,
double valeur, int id_prof) {    this.id_etu =
id_etu;
    this.nom_matiere = nom_matiere;
    this.valeur = valeur;
    this.id_prof = id_prof;
    }
    public int getId_etu(){
    return id_etu; }
    public String getNom_matiere(){
    return nom_matiere; }
    public double getValeur(){
    return valeur; }
    public int getId_prof(){
    return id_prof; }
    public void setId_etu(int id_etu){
    this.id_etu = id_etu; }
    public void setNom_matiere(String m){
    this.nom_matiere = m; }
    public void setValeur(double valeur){
    this.valeur = valeur; }
    public void setId_prof(int id_prof){
    this.id_prof = id_prof; }
}
```

## Script SQL de la Base de Données

```
CREATE TABLE person (  
    id INT AUTO_INCREMENT PRIMARY KEY,  
    nom VARCHAR(100) NOT NULL,  
    prenom VARCHAR(100) NOT NULL,  
    dn DATE,  
    email VARCHAR(150) UNIQUE NOT NULL,  
    role VARCHAR(50) NOT NULL,  
    mtp VARCHAR(100) NOT NULL  
);  
  
CREATE TABLE admin (  
    id_adm INT AUTO_INCREMENT PRIMARY KEY,  
    id_person INT NOT NULL UNIQUE,  
    bureau INT,  
    FOREIGN KEY (id_person) REFERENCES person(id) ON DELETE CASCADE  
);  
  
CREATE TABLE enseignant (  
    id_prof INT AUTO_INCREMENT PRIMARY KEY,  
    id_person INT NOT NULL UNIQUE,  
    speciality VARCHAR(100),  
    FOREIGN KEY (id_person) REFERENCES person(id) ON DELETE CASCADE  
);  
  
CREATE TABLE etudiant (  
    id_etu INT AUTO_INCREMENT PRIMARY KEY,  
    id_person INT NOT NULL UNIQUE,  
    moy DOUBLE DEFAULT 0,  
    niveau VARCHAR(50),  
    FOREIGN KEY (id_person) REFERENCES person(id) ON DELETE CASCADE  
);  
  
CREATE TABLE matiere (  
    nom VARCHAR(100) PRIMARY KEY,  
    coeff DOUBLE DEFAULT 1,  
    id_prof INT,  
    FOREIGN KEY (id_prof) REFERENCES enseignant(id_prof) ON DELETE SET  
NULL  
);  
  
CREATE TABLE note (  
    id_etu INT NOT NULL,  
    nom_matiere VARCHAR(100) NOT NULL,  
    note DOUBLE DEFAULT 0,  
    id_prof INT,  
    PRIMARY KEY (id_etu, nom_matiere),  
    FOREIGN KEY (id_etu) REFERENCES etudiant(id_etu) ON DELETE CASCADE,  
    FOREIGN KEY (nom_matiere) REFERENCES matiere(nom) ON DELETE CASCADE,  
    FOREIGN KEY (id_prof) REFERENCES enseignant(id_prof) ON DELETE SET  
NULL  
);
```

```

INSERT INTO person (id, nom, prenom, dn, email, role, mtp) VALUES
(1, 'Admin', 'User', '1980-01-01', 'admin@school.com', 'admin',
'admin123'),
(2, 'Dupont', 'Jean', '1975-05-10', 'jean@school.com', 'enseignant',
'prof123'),
(3, 'Martin', 'Alice', '2000-03-15', 'alice@school.com', 'etudiant',
'etu123'),
(6, 'Benali', 'Youssef', '2001-07-20', 'youssef@school.com', 'etudiant',
'etu456'),
(7, 'Leclerc', 'Paul', '1978-11-02', 'paul@school.com', 'enseignant',
'prof456'),
(11, 'ali', 'ali', '2001-08-11', 'ali@school.com', 'etudiant', '123'),
(12, 'James', 'Pond', '1970-10-10', 'james@school.com', 'enseignant',
'123');
INSERT INTO admin (id_adm, id_person, bureau) VALUES (1, 1, 101);
INSERT INTO enseignant (id_prof, id_person, speciality) VALUES
(1, 2, 'Informatique'),
(2, 7, 'Maths'),
(4, 12, 'Maths');
INSERT INTO etudiant (id_etu, id_person, moy, niveau) VALUES
(1, 3, 15.5, 'L3'),
(3, 6, 0, 'L3'),
(4, 11, 0, 'L3');
INSERT INTO matiere (nom, coeff, id_prof) VALUES
('Informatique', 5, 1),
('Maths', 1.5, 2),
('physique', 2, 2);
INSERT INTO note (id_etu, nom_matiere, note, id_prof) VALUES
(1, 'Informatique', 19, 1),
(1, 'Maths', 20, 2),
(3, 'Maths', 19, 2),
(4, 'Informatique', 15, 1),
(4, 'physique', 0, 2);

```

!!! Run Screen Recording 2026-05-12 14.mp4 for démonstration for the app !!!!